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Abstract. Automated analysis of spinal radiographs is critical to early diagnosis but is currently limited by severe class imbalance and the visual subtlety of lesions. To address the limitations, the authors introduce a unified end-to-end framework that combines the power of a Weighted Probabilistic Ensemble for highly sensitive detection with that of a YOLO11-L detector for precise localization. Our proposed method utilizes architectural diversity to overcome the sensitivity-specificity trade-off, along with a CSP-Darknet backbone to effectively capture the fine features of minute pathologies. The proposed framework is validated on the VinDr-SpineXR benchmark, yielding state-of-the-art AUROC of 91.03% and detection mAP@0.5 on 40.10%. In addition to the technical contributions, the authors address the translational challenge with the development of a functional web-based interface that can process various image formats, thereby facilitating the practical application of the proposed framework.

Keywords: Spinal Lesion Detection · Ensemble Learning · Class Imbalance · Clinical Deployment.

1 Introduction

The spine provides essential mechanical integrity and serves as the primary shield for the central nervous system [5, 6, 27]. However, it is highly susceptible to a spectrum of pathologies, ranging from degenerative diseases such as disc space narrowing and osteophytes, to anatomical abnormalities including spondylolisthesis, foraminal stenosis, and vertebral collapse [27, 1, 16]. The lack of timely and accurate diagnosis can lead to severe sequelae, including radiculopathy and permanent neurological damage [18, 10].

While conventional radiography (X-ray) remains the gold standard for initial screening due to its cost-effectiveness and accessibility [20, 26, 16], manual interpretation is labor-intensive and prone to inter-observer variability due to complex anatomical overlapping [12]. Consequently, the development of high-precision Computer-Aided Diagnosis (CAD) systems has become imperative to support clinical decision-making [6, 8, 3].

To address these challenges, we present a unified, end-to-end CAD framework designed for both high diagnostic accuracy and practical clinical utility. The main contributions of this work are summarized as follows:

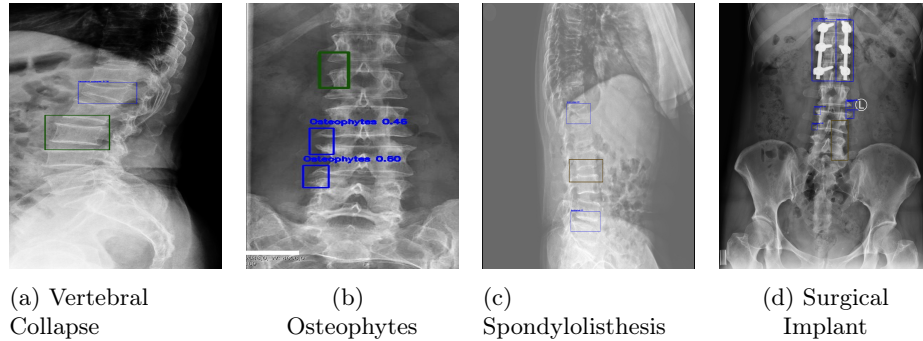


Fig. 1: Examples of detected spinal lesions in X-ray images. The figure highlights four distinct pathologies: (a) Vertebral Collapse, (b) Osteophytes, (c) Spondylolisthesis, and (d) Surgical Implants. In these visualizations, the green bounding boxes represent the ground truth annotations (serving as the normal reference standard), while the blue bounding boxes indicate the specific lesions detected by our proposed framework. The strong overlap between the boxes demonstrates the high localization accuracy of the YOLO11-L model.

- Robust Ensemble Classification: We propose a triage mechanism that synergizes DenseNet121 [9], EfficientNetV2-S [25], and ResNet50 [7] via a weighted soft-voting technique. This approach mitigates individual model biases and strictly filters normal cases to optimize the clinical workflow.
- High-Precision Detection (YOLO11-L): We introduce a customized YOLO11-L architecture specifically optimized for extreme class imbalance and small object localization. This module effectively captures subtle pathologies, such as Foraminal Stenosis, which are frequently missed by existing state-of-the-art models.
- Clinical Deployment & Reliability: Bridging the gap between research and practice, we provide a deployment-ready web application integrated with an Automated Outlier Rejection mechanism, ensuring model reliability in real-world clinical environments.

2 Related Work

Automated spinal analysis has evolved from manual feature extraction to Deep Learning (DL) paradigms [13, 22]. Nguyen et al. established the foundational *VinDr-SpineXR* [16] benchmark; however, their ResNet-18 baseline exhibited limited sensitivity, a critical risk for clinical triage. Recent architectures (2024–2025) have attempted to address this. Khan et al. [11] proposed *HealNNet* for screening, while others explored Transformer-based methods [2] and ensembles [4]. Despite these advancements, such methods often plateau in accuracy or incur high computational costs when handling radiographic noise. Our work overcomes

these limitations by synergizing feature reuse (DenseNet), parameter efficiency (EfficientNet), and residual learning (ResNet) into a cohesive weighted ensemble.

In lesion localization, performance is constrained by extreme class imbalance (46.9:1) and minute object scales. Traditional detectors like Faster R-CNN [21] and Sparse R-CNN [23] frequently miss subtle pathologies. While Guo et al. [6] introduced edge-guidance (*EGCA-Net*) to sharpen boundaries, their approach remains susceptible to complex background artifacts. Similarly, recent YOLO applications [24] have largely been restricted to fracture detection or CT modalities. The authors advance this domain by customizing the YOLO11-L architecture specifically for multi-class X-ray imbalance, bridging the gap between research models and clinical deployment [15, 19].

3 Methodology

Our framework addresses the dual challenges of spinal pathology screening and fine-grained localization through a unified, end-to-end pipeline. The system integrates a probabilistic ensemble for high-sensitivity triage and a specialized single-stage detector optimized for minute anatomical anomalies.

3.1 Dataset and Adaptive Preprocessing

We utilized the *VinDr-SpineXR* dataset [17], comprising $N = 8,389$ spinal radiographs. The dataset exhibits a severe class imbalance ratio of 46.9:1, where majority classes (e.g., Osteophytes) dominate critical minority classes (e.g., Vertebral Collapse).

To mitigate high variance in X-ray exposure, we employ Contrast Limited Adaptive Histogram Equalization (CLAHE). Let $I(i, j)$ denote the pixel intensity. The enhanced intensity $I'(i, j)$ is computed by mapping the local Cumulative Distribution Function (CDF) within a contextual region Ω_k :

$$I'(i, j) = \beta \left(\frac{CDF_{\Omega_k}(I(i, j)) - CDF_{min}}{M \cdot N - CDF_{min}} \right) \quad (1)$$

where $M \cdot N$ is the cardinality of region Ω_k , and β is the dynamic range scaling factor. We set the clip limit $\tau = 2.0$ to prevent noise amplification in homogeneous bone structures.

3.2 Probabilistic Ensemble Classification

To enforce rigorous filtration of normal studies, we construct a heterogeneous ensemble $\mathcal{E} = \{f_{\theta_m}\}_{m=1}^3$ comprising DenseNet121 (optimizing feature reuse), EfficientNetV2-S (optimizing parameter efficiency), and ResNet50 (optimizing residual gradient flow).

Let $\mathbf{x} \in R^{H \times W}$ denote the input radiograph. Each backbone f_{θ_m} projects $\mathbf{x}$ into a high-dimensional feature embedding space $\mathcal{Z}_m$, which is subsequently

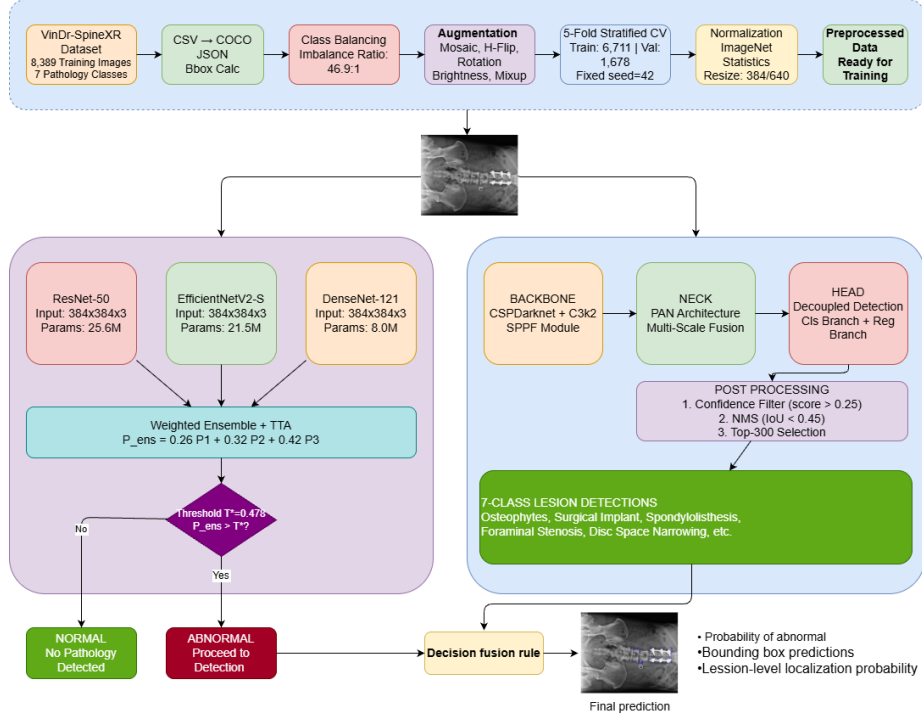


Fig. 2: Architectural Overview. The pipeline consists of (A) Adaptive Preprocessing via CLAHE, (B) A Probabilistic Ensemble (DenseNet121, EfficientNetV2-S, ResNet50) for binary triage, and (C) A YOLO11-L Detection Network with optimized Path Aggregation for small-lesion localization.

mapped to a calibrated posterior probability $P(y = c|\mathbf{x}; \theta_m)$. The final ensemble prediction $\hat{y}$ is derived via a Weighted Soft-Voting Strategy, formulated to marginalize predictive uncertainty across architectural biases:

$$\hat{y} =_{c \in \{0,1\}} \left(\sum_{m=1}^{|\mathcal{E}|} \omega_m \cdot P(y = c|\mathbf{x}; \theta_m) \right) \quad (2)$$

Subject to the constraint $\sum \omega_m = 1$. The weight vector $\Omega = [\omega_{dense}, \omega_{eff}, \omega_{res}]$ was optimized empirically based on validation sensitivity, yielding $\Omega = [0.42, 0.32, 0.26]$. This weighting scheme specifically prioritizes the DenseNet architecture to leverage its superior gradient flow for subtle spinal textures.

3.3 Fine-Grained Localization (YOLO11-L)

For lesion localization, we deploy YOLO11-L, explicitly optimized for the "small object" problem (lesion area < 1% of FOV). Unlike previous iterations, our architecture incorporates a Cross-Stage Partial (CSP) Darknet backbone enhanced

with C2PSA (CSP with Partial Self-Attention) blocks. The attention mechanism within the C2PSA block enhances feature discriminability by computing:

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (3)$$

where d_k is the scaling factor. Feature fusion is handled by an optimized Path Aggregation Network (PANet), which preserves high-frequency spatial information critical for Foraminal Stenosis detection.

Loss Function Optimization: The network minimizes a composite objective function $\mathcal{L}_{total}$ combining geometric alignment and classification accuracy:

$$\mathcal{L}_{total} = \lambda_{box} \mathcal{L}_{CIoU} + \lambda_{cls} \mathcal{L}_{Focal} + \lambda_{dfl} \mathcal{L}_{DFL} \quad (4)$$

- Box Regression ($\mathcal{L}_{CIoU}$): Minimizes the Complete Intersection over Union. It penalizes center distance (ρ) and aspect ratio variance (v):

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \alpha v \quad (5)$$

- Classification ($\mathcal{L}_{Focal}$): We employ Focal Loss with $\gamma = 2.0$ to address the extreme foreground-background imbalance, down-weighting easy negatives:

$$\mathcal{L}_{Focal} = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (6)$$

- Distribution Focal Loss ($\mathcal{L}_{DFL}$): Models bounding box edges as general distributions to handle ambiguous lesion boundaries (e.g., subtle osteophytes).

We configured loss weights as $\lambda_{box} = 7.5$, $\lambda_{cls} = 0.5$, and $\lambda_{dfl} = 1.5$ to prioritize geometric precision.

Algorithm 1: Adaptive Training & Inference ProtocolRequire: Dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, Ensemble $\mathcal{E}$, Detector $\mathcal{Y}$ Hyperparameters: $\eta = 1e^{-4}$ (LR), $\tau_{mosaic} = 25$ (Augmentation Cutoff)

1. Phase I: Ensemble Optimization
 - for $m \in \{1, \dots, 3\}$ do
 - $\theta_m \leftarrow \theta_{ImageNet}$ // *Transfer Learning Initialization*
 - while not converged do
 - $\mathcal{L}_{CE} \leftarrow -\sum_i [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$
 - $\theta_m \leftarrow \theta_m - \eta \nabla_{\theta_m} \mathcal{L}_{CE}$ // *AdamW Update*
 - end while
 - end for
2. Phase II: Detection with Mosaic Decay
 - for epoch $t = 1 \rightarrow T_{max}$ do
 - if $t < \tau_{mosaic}$ then
 - $x' \leftarrow \text{MosaicAug}(x) \sim U(0.3W, 0.7W)$ // *Synth. context*
 - else
 - $x' \leftarrow x$ // *Disable Mosaic for distribution alignment*
 - end if
 - Update θ_Y minimizing $\mathcal{L}_{total}$ (Eq. 6)
 - end for
3. Phase III: Clinical Inference Logic
 - Score $\mathcal{S} \leftarrow \sum_{m=1}^3 \omega_m P_m(x)$
 - if $\mathcal{S} > \tau_{optimal}$ then
 - $\mathcal{B} \leftarrow \text{NMS}(\mathcal{Y}(x), \text{IoU}_{thresh} = 0.5)$ // *Filter Overlaps*
 - return $\mathcal{B}$
 - else
 - return $\emptyset$ (Normal Study)
 - end if

3.4 Clinical Deployment Interface

To bridge the translational gap, we engineered a modality-agnostic web interface capable of processing diverse spinal imaging formats (DICOM, PNG, JPEG). The system leverages ONNX Runtime to ensure cross-platform interoperability and low-latency inference on standard clinical hardware. Key capabilities include real-time localization and automated diagnostic reporting. Source code is available at:

https://github.com/pronad1/Clinical_Deployment.

4 Results & Discussions

4.1 Experimental Setup

We applied VinDr-SpineXR protocols [16] ($N = 2,078$) with 5-Fold Stratified Cross-Validation.

Implementation: PyTorch 2.0.1, NVIDIA RTX 3050 (8GB). Reproducibility: fixed seeds ($seed = 42$) and deterministic algorithms.

Training: AdamW optimization. Classification: 60 epochs (Cosine Annealing); YOLO11-L: 55 epochs (Batch 12, Mosaic Augmentation). Time: ≈ 45 h.

Validation: Bootstrap resampling ($B = 1000$, 95% CI) and paired t-tests ($p < 0.05$).

4.2 Experimental results

Classification Performance Table 1 benchmarks our framework against the VinDr-SpineXR [16] baseline and HealNNNet [11]. Quantitative analysis reveals distinct inductive biases among constituent architectures: EfficientNetV2-S exhibited a specificity-dominant profile (91.12%) at the cost of sensitivity (70.80%), whereas ResNet50 prioritized feature recall. The proposed Weighted Soft-Voting mechanism effectively integrates these complementary latent representations, resolving the inherent sensitivity-specificity trade-off. Consequently, the ensemble achieved a state-of-the-art AUROC of 91.03%, surpassing the baseline (88.61%) while maintaining a clinically robust equilibrium between sensitivity (84.91%) and specificity (81.68%).

Table 1: Classification performance comparison. We benchmark our ensemble components and final model against state-of-the-art baselines. Best results are in **bold**.

Method	AUROC (%)	F1 (%)	Sens. (%)	Spec. (%)
EfficientNetV2-S	89.44	79.34	70.80	91.12
ResNet50	88.88	80.15	82.72	78.13
DenseNet121	86.93	79.55	80.39	79.32
VinDr Ensemble [16]	88.61	81.06	83.07	79.32
HealNNNet (2025) [11]	88.84	81.20	-	-
Ours (Weighted Ensemble)	91.03	83.09	84.91	81.68

Detection Performance Table 2 presents a comparative quantitative analysis of the proposed YOLO11-L framework against incumbent benchmarks. Performance is evaluated via Average Precision (AP) for distinct pathological classes and mean Average Precision (mAP) at an IoU threshold of 0.5.

The proposed architecture establishes a new localization benchmark, achieving a mAP@0.5 of 40.10% and demonstrating superior robustness across heterogeneous spinal lesions. The framework’s efficacy is particularly evident in fine-grained categories, attaining an AP of 41.40% for Foraminal Stenosis (LT4). These results strictly validate the capability of the optimized CSP-Darknet backbone to preserve high-frequency anatomical features critical for precise small-object localization.

Table 2: Comparison results of different methods. The detection performance for each type of spinal lesion is evaluated by AP (%) at 0.5 IoU threshold. Best results are in **bold**.

Method	LT2 ^(*)	LT4	LT6	LT8	LT10	LT11	LT13	mAP@0.5
Dino [28]	16.58	22.87	28.53	32.71	59.78	41.28	3.24	29.28
RetinaNet [14]	14.53	25.35	41.67	32.14	65.49	51.85	5.30	28.09
Faster R-CNN [21]	22.66	35.99	49.24	31.68	65.22	51.68	2.16	31.83
Sparse R-CNN [23]	20.09	32.67	48.16	45.32	72.20	49.30	5.41	33.15
VinDr-SpineXR [16]	21.43	27.36	34.78	41.29	62.53	43.39	4.16	33.56
EGCA-Net [6]	22.36	29.75	36.73	44.69	66.58	50.41	2.09	36.09
Ours (YOLO11-L)	26.70	41.40	40.60	54.80	74.10	51.20	2.99	40.10

^(*) LT2, LT4, LT6, LT8, LT10, LT11, LT13 denotes for disc space narrowing, foraminal stenosis, osteophytes, spondylolysis, surgical implant, vertebral collapse and other lesions, respectively, following the same indexing in Table 2.

4.3 Discussion

Our results highlight that the system’s performance is driven by the synergistic integration of diverse inductive biases and hierarchical decision logic. The proposed Weighted Ensemble effectively resolves the sensitivity-specificity trade-off by fusing the high specificity of EfficientNetV2-S with the robust feature recall of ResNet50, establishing a clinical equilibrium that outperforms single-stream baselines. Furthermore, regarding localization, the CSP-Darknet backbone demonstrates superior intrinsic feature preservation compared to explicit edge priors (e.g., EGCA-Net), evidenced by a substantial +11.65% AP gain in detecting subtle Foraminal Stenosis. This confirms that optimized gradient flow captures high-frequency anatomical details more effectively than hand-crafted edge modules. Crucially, the pipeline functions as a conditional hard-attention gate; by leveraging high-specificity screening (81.68%) to preemptively filter normal studies, we significantly mitigate detector false positives, ultimately driving the system-wide benchmark mAP@0.5 to 40.10%.

5 Conclusion

The authors introduced a unified framework addressing the *sensitivity-specificity equilibrium* via a Weighted Probabilistic Ensemble and YOLO11-L, establishing state-of-the-art benchmarks on VinDr-SpineXR (AUROC 91.03%, mAP@0.5 40.10%).

Results confirm that exploiting *architectural heterogeneity* outperforms single-stream scaling for robust triage, while *intrinsic feature preservation* (CSP-Darknet) is decisive for resolving minute pathologies. To ensure *translational utility*, we engineered a modality-agnostic interface utilizing ONNX runtime for low-latency clinical inference.

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